Rui Xu (许睿)

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EDUCATION

- **Peking University**, School of Physics, Department of Atmospheric and Oceanic Sciences — Beijing, China
  Ph.D. student in Atmospheric Physics and Atmospheric Environment, Advisor: Prof. Yan Yu — Sept. 2024 – Present
- **Peking University**, School of Physics, Department of Atmospheric and Oceanic Sciences — Beijing, China
  B.S. in Atmospheric Science — Sept. 2020 – Jul. 2024

RESEARCH INTERESTS

Atmospheric aerosols, with a specific focus on the emission, transport, and atmospheric processing of wildfire smoke. Integrating satellite remote sensing observations with climate model simulations, I quantitatively assess the spatiotemporal distribution and subsequent transport of wildfire smoke aerosols. I further investigate their impacts on air quality, atmospheric dynamics, and feedback mechanisms within the Earth's climate system.

PUBLICATIONS

[1] Xu, R., Yu, Y., Meng, X., Xue, H., Zhao, C., & Lin, J. (2025). Atmospheric convection and aerosol absorption boost wildfire smoke injection. *Geophysical Research Letters*, 52, e2025GL115989. https://doi.org/10.1029/2025GL115989

[2] Chen, J., Xu, J., Wu, Z., Meng, X., Yu, Y., Ginoux, P., DeMott, P. J., **Xu, R.**, Zhai, L., Yan, Y., Zhao, C., Li, S.-M., Zhu, T., & Hu, M. (2024). Decreased dust particles amplify the cloud cooling effect by regulating cloud ice formation over the Tibetan Plateau. *Science Advances*, 10(37), eado0885. https://doi.org/doi:10.1126/sciadv.ado0885

RESEARCH PROJECTS

1. Modeling of Wildfire Smoke Injection and Vertical Distribution

Developed a physics-based parameterization of smoke injection height and vertical profiles using multi-satellite observations (e.g., MODIS, CALIPSO) and reanalysis meteorology. Improved wildfire representation in climate models by considering aerosol self-lofting and ambient convection processes.

2. Global Impacts of Wildfire Smoke under Climate Change

Detected multi-year trends in the vertical distribution of wildfire smoke from satellite data and new parameterization across key regions. Coupling the newly developed vertical distribution scheme with CESM to investigate climatic effects and underlying feedback mechanisms.

CONFERENCE PRESENTATIONS

- **AGU Annual Meeting 2025** — New Orleans, USA
  Poster: Recently Rising Smoke Plumes over Boreal Forests due to Intensifying Wildfires — Dec. 2025
- **AGU Annual Meeting 2024** — Washington, DC, USA
  Oral: Smoke Vertical Distribution Based on Satellite Remote Sensing and Modeling — Dec. 2024

HONORS & AWARDS

Outstanding Scientific Research Award & Tung Scholarship, *Peking University* — Dec. 2025
Excellent Graduate & Weiming Scholar (top 1% of undergraduate students), *Peking University* — Jun. 2024
Merit Student, *Peking University* — Dec. 2023

PROFESSIONAL SKILLS

Language: Mandarin (native), English (IELTS: 7.5)
Technical Skills: Python, NCL, Linux, CESM

TEACHING EXPERIENCE

Teaching Assistant of General Physics Lab, *School of Physics, Peking University* — Feb. 2025 – Jun. 2025